

Learn-AI Innovation Challenge 2026

Theme: AI for Sustainable Environment & Climate Action

MauBuoy

Coral Reef Health, Prediction, and Restoration

By MARINE DEFENDERS



Sir Abdool Raman Osman State College



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The Crisis: Mauritius' Coral Reefs Are Dying



The Alarming Reality



Rapid Coral Decline

- 40-50% of coral cover lost in the last 20 years
- Sea temperatures rising: +1.2°C since 1980
- Mass bleaching events destroying 80%+ of corals during marine heatwaves



Why It Matters



Economic & Ecological Impact

- Tourism: \$126M/year dependent on healthy reefs
- Fisheries: 1,900+ fishing families at risk
- Coastal protection: Natural barrier against storms & erosion weakening



The Monitoring Gap



Current System Is Failing

- Reactive: Detects damage AFTER it happens
- Manual: Diver surveys are slow, expensive, limited
- Fragmented: Data not shared across agencies

MauBuoy : From Prediction to Prevention

A proactive early-warning system that detects coral stress BEFORE visible bleaching occurs



[Step 1: Collect]

Smart Sensor Network

- Solar-powered buoys at key reef sites
- Through Cables, collects temp, pH, dissolved O₂, turbidity
- Adaptive sampling: increases frequency during heat stress



[Step 2: Analyze]

AI Prediction Engine

- Trained on global coral datasets + Mauritius baseline data
- Forecasts bleaching risk 7–30 days ahead
- Outputs: Risk Score, Disease Probability, Recovery Potential



[Step 3: Act]

Actionable Dashboard

- Real-time alerts to MMCS, fishermen, tourism operators
- Recommends interventions (temporary shading, access limits)
- Tracks intervention impact for continuous AI learning





How MauBuoy Works: Data, Models & Predictions



Combining ocean science with artificial intelligence for accurate coral health forecasting

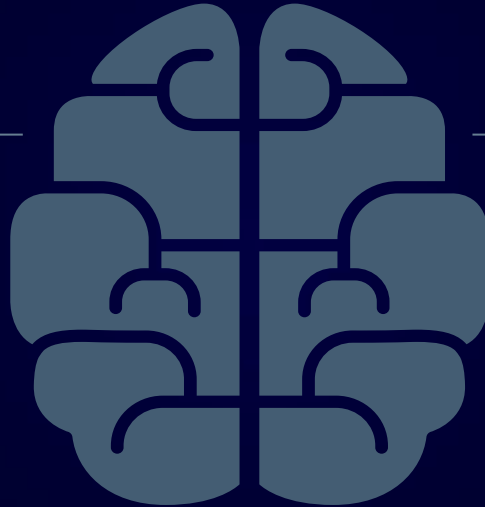
1. Data Strategy: What + Why

STRUCTURED (Numbers)

- Temperature — every 10 min
- pH & Dissolved O₂ — every 30 min
- Turbidity — every 15 min
- Salinity — hourly

UNSTRUCTURED (Images/Text)

- Underwater Photos — daily
- Citizen Reports — on-demand
- Satellite SST — daily



2. Models & Outputs

IMAGE CLASSIFICATION MODEL

Trained on coral health datasets to detect: Healthy, Stressed coral, Bleached, and Diseased corals

PREDICTION MODEL

Analyses sensor data + historical patterns to generate:

1. Bleaching Risk Score (0-100)
2. Disease Outbreak Probability
3. Recovery Potential





Where We Start: Pilot Sites in Mauritius



ALBION COAST

- Why: High wave exposure + bleaching vulnerability
- Sensor depth: 5-20m



POINTE-AUX-FEUILLES

- Why: Biodiversity hotspot + high tourism pressure
- Sensor depth: 2-10m



BLUE BAY MARINE PARK

- Why: Active coral restoration site
- Sensor depth: 3-15m



MauBuoy Dashboard: Your Reef Health Forecast



Real-time alerts + actionable insights for smarter conservation

Dashboard Mockup Visual

CURRENT STATUS: BLUE BAY MARINE PARK

Risk Score: 68/100 (Moderate → High)

30-DAY PREDICTION

- Days 1-7:  Moderate risk (65%)
- Days 8-21:  High risk if temp >30.2°C
- Days 22-30:  Recovery possible with action

INTERVENTION IMPACT

"If temporary shading applied within 48h:
→ 75% probability of recovery in 45 days"

Actionable For



MMCS Scientists

- Early-warning alerts
- Data for policy decisions



Fishermen & Coastal Communities

- SMS alerts in Creole/French
- Safe fishing zone updates



Tourism Operators

- Real-time reef health info for guests
- "Climate-resilient tourism" marketing



Ministry of Environment

- National reef monitoring dashboard
- SDG 13 & 14 progress tracking



Responsible AI: Our Ethics & Governance Framework



Building trust through transparency, privacy, and human oversight

Our Core Principles



Data Privacy First



Human-in-the-loop



Transparency & Explainability



How We Protect Everyone



Bias Mitigation



Accessibility & Inclusion



Continuous Improvement



AI + Humans: Stronger Together



MauBuoy amplifies human expertise — it doesn't replace it

AI's Role

- 24/7 monitoring of temperature, pH, oxygen
- Detects subtle patterns humans might miss
- Forecasts bleaching risk 7-30 days ahead
- Sends instant alerts to stakeholders



Collaboration



Human's Role

- Validate AI predictions with local knowledge
- Make ethical decisions on interventions
- Provide feedback to improve AI accuracy



FEEDBACK LOOP: Action → Data → Learning → Better AI



From Idea to Impact: Our Implementation Roadmap

A phased approach built for Mauritius

- Add Pointe-aux-Feuilles + Albion sites
- Launch citizen science mobile app for report submissions
- Integrate with MMCS alert system

Pilot Expansion

01

Prototype

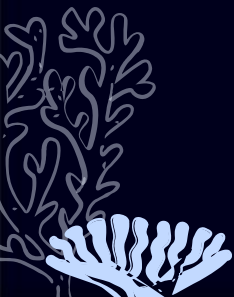
- Deploy 3 solar-powered sensor buoys in Blue Bay
- Train AI on global coral datasets + local baseline data
- Build mock dashboard for MMCS testing

02

03

National Scale

- Cover all major reef zones in Mauritius
- Export model to other islands
- Long-term: Self-sustaining through public-private partnerships





Let's Protect Mauritius' Reefs - Together



With MauBuoy, prediction becomes prevention.

Core Value Proposition

- ✓ One dashboard. Many users. One mission.
- ✓ AI provides intelligence. Humans provide wisdom.
- ✓ Built for Mauritius. Designed to scale. Ready to start.

Next Steps

- Partner support for Phase 1 prototype (Blue Bay)
- Access to MMCS baseline reef data
- Mentorship from AI + marine science experts
- Opportunity to pilot with coastal schools.



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THANK YOU! – QA Session

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